

EduWand

AI Module — Build Document

Field	Detail
Document	EduWand AI Module — Build Document
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Scope	The AI module only. The core application, data layer, authentication, roster and school setup are already in place and are not covered here.

HOW TO USE THIS Sections 2 to 7 define what to build. Each module states its aim, its workflow, its functional requirements, what 'done' looks like, and the edge cases to handle. Section 8 lists the open questions that still need answers — raise anything else you find there rather than assuming.

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1. Module overview

The AI module consists of six components, with Assignment Lab specified below in two parts — generation and evaluation. They are interdependent: work generated in Lesson Studio feeds Assignment Lab, the Attainment Report and the Communication Hub, so the data produced by each must be reusable by the others rather than siloed.

#	Component	One-line purpose
1	Lesson Studio	Generate curriculum-aligned teaching materials for a topic.
2	Assignment Lab — generation	Generate, verify and distribute assignments to a class or a student.
3	Assignment Lab — evaluation	Grade submissions against the teacher's answer key and produce class-level insight.
4	Attainment Report	A standard, printable per-topic record of objectives, observations and outcomes.
5	Students Dashboard	Student interface for accessing materials, submitting assignments and viewing results.
6	Student Analytics Dashboard	One page for the teacher to see a student's or class's report.
7	Communication Hub	Centralised teacher-to-parent, teacher-to-student and teacher-to-class communication.

Note: the summary list in the original brief named five modules; the body described six. Both the Students Dashboard and the Student Analytics Dashboard are in scope and are listed separately above.

1.1 Terms used in this document

These terms carry a fixed meaning throughout. Use the same names in code, database and UI.

Term	Meaning
Topic	The container for all work on a unit of teaching. Every generation, assignment and attainment report belongs to exactly one topic. A teacher either starts a new topic or continues an existing one.
Context	Material supplied to a generation: an uploaded file, a URL, or an item pulled from the iDream K12 content library.
Generation	A single AI-produced output — lesson plan, activity, flashcard set or presentation — belonging to a topic.
Observation	A note recorded by the teacher while or after conducting the class, attached to the topic and used in the attainment report.
Answer Key	The correct answers and marking guidance produced alongside an assignment. The teacher-verified version is authoritative, never the AI's original.
Plan mode / Generate mode	Plan mode is when a teacher wants to see what is going to be generated, which gives them the flexibility to edit the outline of the output
School format	The output template applied to all generations so that documents are standardised across the institution.

2. Lesson Studio

2.1 Aim

- Help teachers generate curriculum-aligned learning materials to conduct classes (CBSE / ICSE / IB).
- Produce outcome-based and NEP-aligned materials.
- Follow structured pedagogical methods — Bloom's taxonomy — in producing the outputs.

2.2 Workflow

1. Teacher logs into the application and select/filter the class
2. The teacher selects either to start a new topic or to continue a topic worked on previously.
3. On starting a new topic, the teacher is taken to the content library to see existing material on the topic (iDream K12 Content API).
4. The teacher can either refer directly to the iDream K12 content library, or upload their own context for what is being taught. This may be a PDF, Word document, presentation, photograph or a URL.
5. Once context is loaded, the teacher moves to the generation page and sets the class context: class, subject, topic, language, and where to search for content. This page requires no prompting by the teacher — it is a dropdown-driven dashboard built for easy navigation.
6. The teacher selects the output type: lesson plan, custom activity report, flashcards or presentation.
7. Each generation is saved under a topic — the same topic selected at the first step on login.
8. Every output is produced in the school-given format, so that there is standardisation across the institution.
9. The teacher then specifies how many classes this will cover, how much time per class is expected, whether to proceed in plan mode or generate mode, and any custom prompts to include in the generation.
10. The output is generated and automatically saved under the teacher's account. The teacher can send it to students and parents.
11. The generation is also available for weekly automated updates to parents covering what was taught in class.
12. The generation flows into the attainment report, covering what was done, what the objectives were, what the outcomes were, and what can be improved next time.
13. The teacher can capture observations while conducting the class; these also feed the attainment report.

2.3 Functional requirements

- Topic selection on entry, listing the teacher's existing topics and allowing creation of a new one.
- Context ingestion supporting PDF, DOCX, PPTX, images and URLs, plus selection from the iDream K12 content library.
- Generation setup page driven entirely by dropdowns and numeric inputs. No prompt writing is required of the teacher at any point.
- A custom prompt field for teachers who want to add specific instructions. Optional.
- Output types: lesson plan, custom activity report, flashcards, presentation, explanatory videos..
- Class count and per-class duration inputs, which shape the structure of the generated output.
- Plan mode and generate mode selection.
- All output rendered in the school format template.
- Auto-save of every generation under its topic and the teacher's account.
- Teacher must be able to review and edit the generated output before saving or sharing. The edited version is what persists.
- Distribution to students and to parents.
- Weekly automated parent update assembled from the topic's generations.
- Observation capture against the topic — a lightweight note the teacher can add during or immediately after class.

- Automatic feed of generation data and observations into the attainment report, with no re-entry by the teacher.

Acceptance criteria

#	Criterion
1	A teacher can go from login to a saved, school-formatted lesson plan without writing a prompt at any step.
2	Every generation is retrievable under its topic, with the context sources used still attached.
3	An edited generation is what saves, what shares and what flows into the attainment report — not the original AI output.
4	The same topic can be returned to across multiple sessions and days, with all prior generations intact.
5	All four output types render correctly in the school format template.
6	Observations recorded during class appear in the attainment report for that topic without further action.
7	A generation that fails is surfaced to the teacher with the inputs preserved and a one-click retry.

Edge cases to handle

- Uploaded PDF is a scan with no extractable text.
- Uploaded file is in a different language from the selected output language.
- URL is dead, paywalled, or returns a page that requires JavaScript to render content.
- iDREAM K12 library returns no results for the topic.
- Teacher requests an unusually large generation — for example 20 classes of 90 minutes. Define and enforce a ceiling.
- Two topics with the same name for the same class and subject.
- Generated content exceeds the space the school format template allows.
- Teacher loses connection mid-generation.
- Teacher shares a generation to parents before it has been edited.

3. Assignment Lab

3.1 Aim and intended behaviour

Help teachers create assignments to evaluate the understanding of the class and of individual students.

The intent is that drafting an assignment and sharing it with the class becomes a daily habit for the teacher. This is a daily assignment tool. It does not replace the systematic examinations a school already runs.

In the intended flow the teacher logs in, adds sources and context — either preloaded from Lesson Studio or a fresh source such as a PPT, PDF, report or URL — selects the format and goal of the assignment, checks the generated questions and answer key, and distributes to the class. On submission the teacher receives an evaluated assignment with feedback, a summary, and next steps as actions against individual responses.

The module divides into two parts, specified separately below:

- Teacher generation module (Section 3.2)
- Backend AI evaluation and grading (Section 3.4)

3.2 Teacher generation module

Aim: help teachers generate assignments for a class or for an individual student.

Workflow

14. Teacher logs in and navigates to Assignment Lab.
15. To feed context, the teacher either refers to generations from Lesson Studio or uploads fresh material.
16. The teacher sets the assignment format: number of questions, type of questions, difficulty level split, due date, and custom instructions.
17. The teacher specifies which answers require a photo submission and which are online submissions.
18. The assignment is generated along with an answer key, which the teacher checks first.
19. The teacher can change anything they do not like before pushing it to students.
20. The teacher distributes the assignment — immediately, or scheduled for a later day.

Functional requirements

- Context selection from existing Lesson Studio generations, or fresh upload.
- Format controls: question count, question types, difficulty split, due date, custom instructions.
- Per-question flag for photo submission versus online submission.
- Generation of questions with a matching answer key.
- Teacher verification step before distribution. The teacher can edit any question, any answer and any mark allocation.
- Distribution to a whole class or to a named individual student.
- Scheduling: distribute now, or at a specified future date and time.

3.3 Personalisation

Prerequisite: to personalise an assignment, the teacher must have already distributed and graded a minimum of two assignments on the same topic. This is what makes it possible to understand what the student has understood and where they are lagging.

Method 1 — assignment for one specific student

- The teacher generates an assignment for a single named student.
- That student's performance is loaded to personalise the assessment.
- Performance means their assignment responses and the teacher's feedback. Nothing else.

Method 2 — personalisation at class level

- The teacher activates the personalisation toggle. It is off by default.
- The toggle adds 2 - 5 questions on top of the generated assignment, per student.
- These questions reflect what that student needs to work on next, needs to revisit, or needs to be re-tested on for that topic.
- This separate section is based entirely on how the student performed in previously generated assignments.

BUILD NOTE Class-level personalisation means the system produces a distinct question set per student, not one set per class. For a class of 40 that is 40 generations rather than 1.

Acceptance criteria

#	Criterion
1	A teacher can go from topic to distributed assignment without leaving Assignment Lab.
2	No assignment reaches a student without an explicit teacher confirmation action.
3	A teacher edit to a question, an answer or a mark allocation persists and is what both the student and the grading engine see.
4	The two-graded-assignments prerequisite for personalisation is enforced server-side, not only hidden in the UI.
5	Where the prerequisite is not met, the teacher is told plainly why personalisation is unavailable and the standard assignment still generates.
6	A scheduled assignment distributes reliably at the scheduled time.
7	Photo-submission and online-submission flags carry through correctly to the Students Dashboard.

Edge cases to handle

- A student joins the class mid-topic and has no prior graded assignments.
- A student was absent for one or both of the prior assignments.
- The teacher deletes or archives a topic that has distributed assignments attached.
- Due date is set in the past.
- The teacher edits an assignment after it has been distributed but before students have submitted.
- A student submits after the due date.
- Personalisation toggle is switched on for a class where only some students meet the prerequisite.

3.4 Backend evaluation and grading

Aim: when a student submits an assignment, the response is checked and graded against the answer key put in place by the teacher at generation.

What the evaluation must tell the teacher

Whether the class has understood the topic, by segregating and identifying performers across the class into three levels:

Level	Band (editable)	Meaning
Level 1	Above 80%	Top performers.
Level 2	50% to 80%	Middle band.
Level 3	Below 50%	Students needing more attention.

- Who the top performers are and which students need to be given more attention.
- Whether the class as a whole has understood the topic, and how difficult that concept proved to be.
- Which questions were answered incorrectly by the majority of the class.
- What activities or actions can be done in class to help the children improve their understanding of the concept.

Functional requirements

- Grading of submitted responses against the teacher-verified answer key.
- Photo responses are extracted using OCR and then graded; online responses are graded directly from the submitted data.
- Per-student feedback, summary and next-step actions against individual responses.
- Class banding into the three levels above, with named students in each band.
- Item analysis identifying questions the majority answered incorrectly.
- Suggested classroom actions to address the concepts the class struggled with.
- Results feed the Attainment Report outcomes and the Student Analytics Dashboard.
- Teacher can override any grade. The override is recorded as the authoritative value.

Acceptance criteria

#	Criterion
1	Objective question types grade deterministically and match the teacher's answer key exactly.
2	Every graded submission carries a grade, feedback and a next step.
3	Class banding is correct and reproducible against a known test set.
4	A teacher override replaces the AI grade everywhere it appears, including in the attainment report and analytics.
5	Grades become visible to the student only once the teacher releases them.
6	OCR accuracy is measured on a real sample of handwritten submissions and reported before the feature is enabled for a live class.

Edge cases to handle

- Illegible or partially legible handwriting.
- Photo is blurred, rotated, cropped, or poorly lit.
- Answer written in a script or language the OCR has not been tested against.
- Mathematical notation, working, or diagrams within a photo answer.

- Student uploads one photo covering several questions, or the wrong photo entirely.
- A correct answer expressed differently from the wording in the answer key.
- Student submits nothing, or submits a blank response.
- Late submission arriving after the rest of the class has been graded.

RISK Handwriting OCR across Indian school scripts and mathematical notation is the highest-uncertainty piece in this module. A wrongly graded child reaches the parent quickly. AI grade on photo submissions is presented to the teacher as a suggestion for confirmation rather than published directly.

4. Attainment Report

4.1 Aim

A detailed report covering what was done in the class, what the objectives were, what the observations were as recorded while conducting the class, and what the outcomes were.

It is a standard document, replicable across the school, that helps with audit and compliance. It is structured on Bloom's taxonomy and integrated into the application. It is a different form of the analytics dashboard — printable, shareable, and usable in compliance and audits.

4.2 Functional requirements

- One report per topic, populated automatically from Lesson Studio generations, teacher observations and Assignment Lab results.
- Sections: what was done; objectives; observations recorded during class; outcomes; what can be improved next time.
- Structured against Bloom's taxonomy levels.
- Identical structure across every teacher and every class in the school.
- Printable and shareable. Export to PDF, in the school format.
- Roll-up across a class and subject for a term, exportable as a single document.
- Visibility rules: a teacher sees their own reports; school leadership sees all.

Acceptance criteria

#	Criterion
1	A completed topic produces a full attainment report with no additional data entry by the teacher.
2	Objectives and outcomes shown are the ones attached to the topic, not free text re-entered by the teacher.
3	Bloom's taxonomy levels are visibly labelled against objectives.
4	The report prints cleanly to PDF in the school format, with no clipped or overflowing content.
5	A term of reports for one class and subject exports as a single document.

Edge cases to handle

- Topic was generated but never taught — no observations and no outcomes. The report must show the gap honestly rather than fabricate content.
- The teacher heavily edited the generated plan; the report must reflect what was actually delivered.
- Topic spans a term or academic-year boundary.
- No assignment was ever set for the topic, so there are no attainment results to report.

5. Students Dashboard

5.1 Aim

The student interface where students access learning materials generated by the teacher, view and submit the assignments assigned to them, and interact with or reach out to their teacher. In phase 2 an AI Socratic chatbot will be added to support them through the assignment.

5.2 Workflow

21. The student answers questions directly in the application, or uploads a photo for questions that require it.
22. Answers submitted online have the data sent directly. Photo answers are scraped using OCR and then graded using AI.
23. Once the teacher has graded the assignment, the student can see their submission and its result.
24. All submitted assignments remain available to the student to review.

5.3 Functional requirements

- View learning materials shared by the teacher.
- View assigned assignments with due dates and status.
- Answer online where the question type allows; upload a photo where the question requires it.
- Submit, with clear confirmation of submission.
- View graded submissions and teacher feedback once released.
- Full history of past submissions, available to review at any time.
- Reach out to the teacher — routed through the Communication Hub.

Phase 2, not in this build: AI Socratic chatbot to support the student through the assignment.

Acceptance criteria

#	Criterion
1	A student can complete and submit an assignment on a mobile browser without assistance.
2	A part-completed attempt survives the browser closing or the connection dropping, and can be resumed.
3	Photo upload succeeds on a slow connection with a typical phone-camera image.
4	Grades and feedback are visible only after the teacher has released them.
5	Past submissions remain accessible after the topic has moved on.

Edge cases to handle

- Student submits after the due date.
- Student uploads a photo of the wrong page or the wrong question.
- Student opens the same assignment on two devices at once.
- Photo upload fails partway; the rest of the submission must not be lost.
- Student has no device access at home.

6. Student Analytics Dashboard

6.1 Aim

For the teacher to have one page on which to see a student's or a class's report in one place.

6.2 What is established so far

- One page, teacher-facing.
- Two levels of view: individual student, and whole class.
- Draws on the same underlying data as the Attainment Report — Lesson Studio generations, observations, and Assignment Lab results.
- Distinguished from the Attainment Report in that the Attainment Report is the printable, shareable, audit-oriented form; this is the interactive form.

7. Communication Hub

7.1 Aim

Centralise the communication channel through the application between parent, teacher and student.

7.2 Channels in scope

Channel	Purpose
Teacher to parents	Automatic weekly updates covering what was done in class and how their child performed, assembled from Lesson Studio and Assignment Lab data.
Teacher to student	Communication between teacher and an individual student regarding doubts or personal reminders.
Teacher to class	Class notices and reminders sent to the whole class at once.

7.3 Functional requirements

- Weekly parent update generated automatically from the week's Lesson Studio generations and Assignment Lab results, per child.
- One-to-one teacher-to-student messaging.
- Teacher-to-class broadcast.
- Message history retained and viewable by the participants.
- Delivery mechanism to be confirmed — see Section 8, Q-08.

Acceptance criteria

#	Criterion
1	A parent with a child in the class receives one weekly update covering that child, on schedule.
2	A parent with more than one child in the school receives one update per child, not one merged message.
3	A teacher-to-class broadcast reaches every student in the class and no student outside it.
4	Message history persists and is viewable by both parties.
5	A week with no teaching activity does not produce an empty update.

Edge cases to handle

- Parent contact details missing or invalid.
- Parent linked to children in several different classes.
- Student moves class mid-term.
- Weekly update falls in a holiday or examination week.
- Teacher wants to review or stop an automatic update before it sends.

8. Next steps — open questions

These are not scope changes. They are decisions and details that the modules above depend on and that the brief does not yet answer. Each needs an owner and an answer before or during build.

8.1 AI layer

ID	Question	Why it matters
Q-02	Is generation synchronous or queued? A multi-class lesson plan may take 60–120 seconds.	A synchronous request/response design will time out under real use and be rebuilt. Needs deciding before architecture is fixed.
Q-03	What is the acceptable cost ceiling per generation, and is there a cap per teacher or per school?	Class-level personalisation produces one generation per student. For a class of 40 that is 40x the cost and the time of a single generation.
Q-07	What exactly does the Student Analytics Dashboard show, and how does it differ from the Attainment Report?	One line of spec today. Cannot be estimated as it stands.
Q-10	What happens when a generation fails or produces poor output? Retry, regenerate, fall back?	Undefined today. It will happen daily in production.
Q-11	Are there guardrails on generated content for factual accuracy against the supplied source, and for age-appropriateness?	A wrong fact in a lesson plan a parent sees is a reputational problem, not a bug.
Q-12	Are generations versioned — prompt version and model version stored against the output?	Attainment reports are used for audit. Outputs should be explainable after the fact and reproducible when the model changes.

8.2 Curriculum and content

ID	Question	Why it matters
Q-13	Where does the syllabus mapping come from — chapter and topic structure, and learning outcomes per class and subject?	'Curriculum-aligned' and the Bloom's-structured attainment report both depend on it. Candidate sources: the iDream K12 taxonomy if they expose it; the DIKSHA / Sunbird framework API; NCERT learning outcome documents; CBSE Academic curriculum documents.
Q-16	Who supplies and maintains the school format templates — the school, EduWand at onboarding, or both?	Schools provides data based on the template

8.3 Data protection

ID	Question	Why it matters
Q-17	How is parental consent captured and recorded for student data?	Section 9 of the DPDP Act 2023 and Rule 10 of the DPDP Rules 2025 require verifiable parental consent for processing a child's data. Schools have a narrowed carve-out for educational tracking; whether that extends to a platform operating behind the school is not settled. Verify with counsel.
Q-18	Is customer data excluded from model training by default at the provider level?	Should be off by default and contractually excluded, with per-school opt-in only.

ID	Question	Why it matters
Q-19	What is the retention and deletion policy for student submissions, generations and analytics?	1 year

End of document. Raise anything unclear before building rather than assuming an interpretation.